

MODULE 7

Load Economics & Lane Planning

Know what a mile costs before you say yes to what it pays.

Cost per mile, fixed and variable costs, diesel and fuel surcharges, break-even and minimum rates, market rate tools, load-to-truck ratios, seasons, rate per day, and planning a whole week for Truck 12.

\$2.59

Truck 12 break-even per mile

3

week plans to compare

~6 hours

study + practice

1

worksheet you'll use daily

How to use this module

In Module 6 you screened loads against Mike's minimum of **\$2.40 per total mile**. Where did that number come from, and is it still right? This module teaches you to **build the number yourself** and to judge a whole week, not a single load.

Truck 12's money facts (practice)

Blue Line Transport · Truck 12

Fixed costs **\$1,300 a week** (truck + trailer payments, insurance, ELD, phone, board seat) · driver Carlos paid **\$0.60 a mile** · **6.5 mpg** · repairs, tyres, tolls **\$0.26 a mile** · runs about **2,500 total miles a week** · dispatch fee **6%** · factoring **3%**.

These are practice numbers for learning. A real carrier gives you its own, or you build them together (Lesson 7.1).

Study rhythm (about 6 hours)

- **Session 1 (1.5 h):** cost per mile, fixed vs variable, fuel.
- **Session 2 (1.5 h):** break-even, minimum rate, the rate con.
- **Session 3 (1 h):** rate tools, load-to-truck, seasons.
- **Session 4 (2 h):** rate per day, the week plan, drill 9, practice.



Bring a calculator

Every page has numbers. Do the math yourself before you read the answer. Speed comes from repetition.

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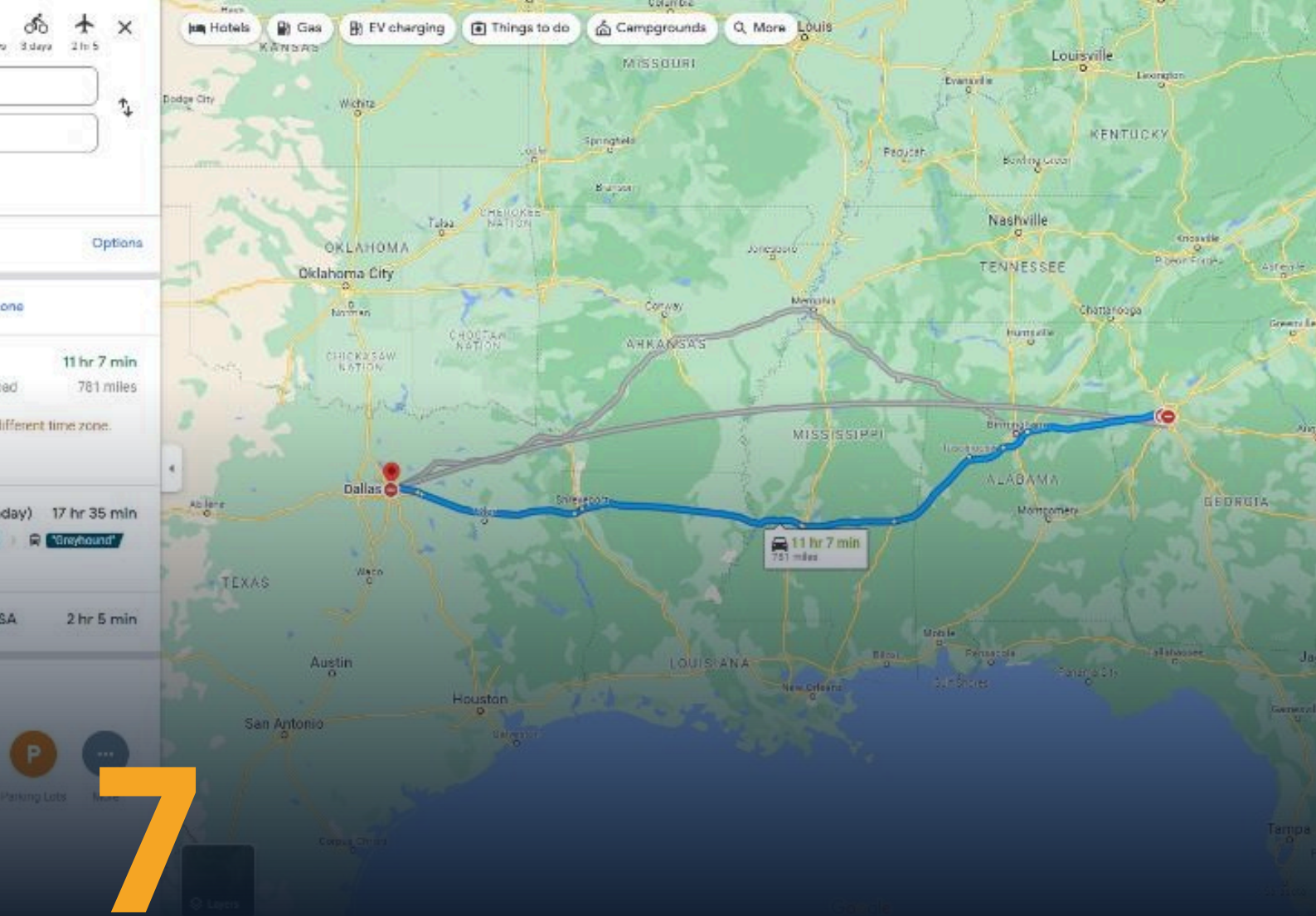
Numbers expire

Diesel, rates and market conditions change weekly. This handbook uses September 2026 figures to **teach the method**. Before using any number with a real carrier, check today's value from the source named on the page.



Carrier-side, always

You work for the carrier. The goal is the carrier's profit per week, not the highest number on one rate con.



MODULE 7 · LOAD ECONOMICS

“Is \$2,050 a good rate?” It depends. On what, exactly?

On what the truck costs to run, what diesel costs this week, what the lane usually pays, how many days the load takes, and where it leaves the truck. By the end of this module you'll answer with numbers, not feelings.

Where the money goes

Where every mile's money goes

Cost per mile = all costs ÷ all miles (loaded + empty). Truck 12 uses practice numbers at September 2026 diesel; ATRI is the U.S. industry average for 2025.

Truck 12 (practice) · 2,500 total miles a week · diesel \$6.38 · 6.5 mpg



ATRI 2025 industry average · real survey data · fuel at 2025 prices



Fuel is the swing cost. ATRI's fuel line was about \$0.48 at 2025 diesel prices. At \$6.38 diesel, Truck 12 pays \$0.98 a mile, twice as much.

Not on the bar yet: percentage fees. Dispatch (6%) and factoring (3%) are taken from the **rate**, so they are added in the break-even step.

Cost per mile, in one line

Cost per mile = all costs for a period ÷ all miles in that period, loaded and empty. If Truck 12 spends \$5,900 in a week and runs 2,500 miles, it costs **\$2.36 a mile**.

Use **total miles**: an empty mile burns the same fuel and pays the same driver as a loaded one.



ATRI benchmark (real data)

The American Transportation Research Institute surveys carriers every year. For 2025 it reported an average **\$2.336 per mile**, a record, and **\$1.854** without fuel. Driver wages were 81.8¢ and repairs 21.5¢ a mile. Use it to sanity-check a carrier's numbers, not as their numbers.



Correction: "Every carrier costs about \$2 a mile"

Cost depends on the truck payment, the driver pay, insurance, fuel price and miles run. Two trucks on the same lane can differ by 50¢ a mile. Always use the carrier's own numbers.

Truck 12's cost sheet

Fixed (per week)	\$ / week	\$ / mile*
Truck payment	600	0.24
Trailer payment	200	0.08
Insurance (liability, cargo, physical damage)	350	0.14
ELD, phone, board seat, permits, accounting	150	0.06
Total fixed	1,300	0.52

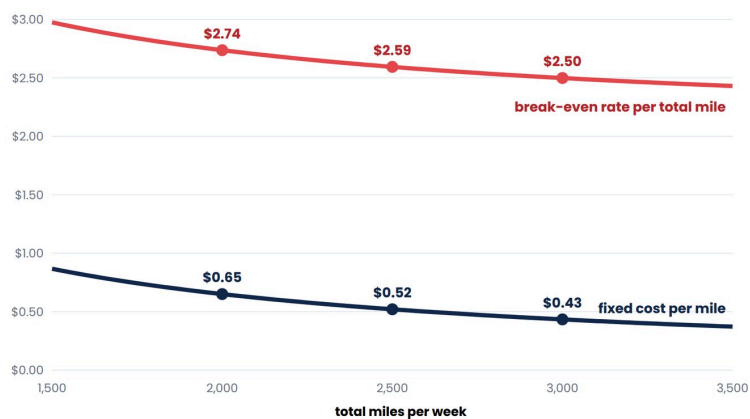
Variable (per mile)	\$ / mile
Fuel (\$6.38 ÷ 6.5 mpg)	0.98
Driver pay (Carlos)	0.60
Repairs & maintenance	0.18
Tyres	0.05
Tolls & misc.	0.03
Total variable	1.84

*At 2,500 total miles a week.

Operating cost = \$0.52 + \$1.84 = \$2.36 a mile.

Fixed costs shrink per mile as the truck runs more

Truck 12 practice numbers: \$1,300 fixed per week, \$1.84 variable per mile, 9% fees.



Fixed costs are paid every week even if the truck sits: truck and trailer payments, insurance, ELD, phone, load-board seat. Truck 12: **\$1,300 a week**.

Variable costs grow with each mile: fuel, driver pay per mile, repairs, tires, tolls. Truck 12: **\$1.84 a mile**.

More miles, lower cost per mile. At 2,000 miles the truck needs \$2.74 to break even; at 3,000 miles, \$2.50. A truck that sits two days loses money even on "good" loads.



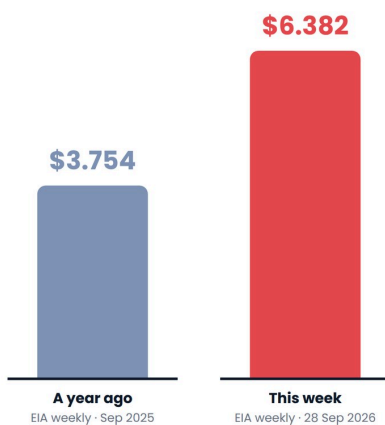
Idle days are expensive

Fixed costs don't stop on weekends or when a truck waits for a load. Keeping the truck moving lowers the cost of every mile. That's why rate per day matters (Lesson 7.5).

Fuel: the swing cost

Diesel: the number that moves your break-even

U.S. average on-highway diesel, EIA. Fuel per mile = price per gallon ÷ miles per gallon.



Truck 12	\$3.75 diesel	\$6.38 diesel
Diesel price	\$3.75	\$6.38
Fuel per mile (÷ 6.5 mpg)	\$0.58	\$0.98
Fuel for a 2,500-mile week	\$1,444	\$2,455
Truck 12 break-even per total mile	\$2.15	\$2.59

Same truck, same loads: +\$1,011 a week in fuel. A minimum rate set last year can now lose money. Recalculate when diesel moves.

Check the price every Monday: EIA "Gasoline and Diesel Fuel Update" (eia.gov) publishes the U.S. average on-highway diesel price weekly. Regional prices differ.

Fuel per mile

Fuel cost per mile = diesel price ÷ mpg. A heavy 53' van usually gets about 6–7 mpg. Hills, speed, weight, weather and idling change it. The carrier knows its real number from fuel receipts and the ELD or fuel card.

Each **\$1.00** rise in diesel adds about **15¢ a mile** at 6.5 mpg ($1 \div 6.5 = \$0.154$). On 2,500 miles that's about **\$385 a week**.



Where to check

EIA Gasoline and Diesel Fuel Update, published every week for the U.S. and by region. Week of 28 Sep 2026: **\$6.382**, against **\$3.754** a year earlier.



Pakistan tip

Put the EIA page in your Monday routine, just like the load board. When diesel moves more than about 25¢, tell the carrier and re-check the minimum rate together.

How fuel gets paid

Fuel surcharge: how fuel is paid separately

DAT's RateView method: (current EIA price - \$1.25 base) ÷ MPG. Worked with September 2026 diesel.

$$\left(\begin{array}{c} \text{This week's EIA diesel} \\ \$6.382 \end{array} - \begin{array}{c} \text{Base price (DAT RateView)} \\ \$1.25 \end{array} \right) \div \begin{array}{c} \text{Miles per gallon (van)} \\ 6.0 \end{array} = \begin{array}{c} \text{Fuel surcharge} \\ \$0.86 / \text{mi} \end{array}$$

On a 760-mile load

760 × \$0.855 = **\$650** of the pay is fuel surcharge.
DAT's national August 2026 average van surcharge was 70¢ a mile; diesel rose in September.

Two ways a broker quotes

All-in: "\$2,050 total." Fuel is inside. Most spot loads.
Linehaul + FSC: "\$1,400 plus fuel." Ask how the surcharge is calculated and get it on the rate con.

Formulas differ. Each shipper or broker uses its own base price and mpg. DAT's \$1.25 base is just one common benchmark. Always compare the **all-in total**.

Worked example

A broker offers "**\$1,400 linehaul plus fuel**" for 700 loaded miles and uses DAT's method: $(\$6.382 - \$1.25) \div 6.0 = \$0.855$ a mile.

Fuel = $700 \times \$0.855 = \599 . All-in = $\$1,400 + \$599 = \$1,999$ = \$2.86 per loaded mile. If deadhead is 50 miles: $\$1,999 \div 750 = \2.67 per total mile.



Always convert to all-in

Compare loads on the **all-in total ÷ total miles**. Two brokers can say "\$1,400" and mean very different money.



Get it in writing

If the rate is "linehaul + FSC", the rate confirmation must show how fuel is paid: a fixed amount, or the formula and the diesel price week. Otherwise the carrier may be paid only the linehaul.

What a rate can include

Part	Meaning	Module
Linehaul	Pay for moving the load from A to B, without fuel	This module
Fuel surcharge	Extra pay that follows the diesel price	This module
Accessorials	Detention, layover, lumpers, extra stops, TONU	Module 10
Deductions	Money taken off for broken rules (tracking, late POD)	Next pages · Module 9

Break-even and minimum rate

From cost per mile to a minimum rate

Truck 12 practice numbers, September 2026 diesel. All rates are per TOTAL mile (loaded + deadhead).



THE FORMULA

Break-even = cost per mile ÷ (1 - fee %)

Divide, don't add: fees are a % of the rate, and the rate is what you're solving for.

Check: $\$2.59 \times 800 \text{ mi} = \$2,072$. Fees 9% = \$186.
 $\$2,072 - \$186 = \$1,886 = 800 \times \2.36 ✓

Mike's old minimum was \$2.40. He set it when diesel was about \$5. At \$6.38 it is below break-even: every mile at \$2.40 loses about 19¢.

Three numbers every dispatcher knows

Number	Truck 12
Operating cost	\$2.36 per total mile
Break-even	\$2.59: below this, every mile loses money
Target minimum	\$2.70: break-even plus the carrier's profit cushion



Why not just add 9%?

$\$2.36 + 9\% = \2.57 , which is too low. The fees are 9% of the **rate**, not of the cost. At \$2.57 the fees are 23¢ and the carrier keeps \$2.34, less than cost. Dividing by 0.91 fixes it.



The carrier sets the minimum

You calculate and explain. Mike decides his profit cushion and when to accept less for a good reason.

Truck 12 break-even at different diesel prices and miles

Diesel (\$/gal)	Fuel / mi	2,000 mi / week	2,500 mi / week	3,000 mi / week
\$4.50	\$0.69	\$2.42	\$2.27	\$2.18
\$5.00	\$0.77	\$2.51	\$2.36	\$2.27
\$5.50	\$0.85	\$2.59	\$2.45	\$2.36
\$6.00	\$0.92	\$2.67	\$2.53	\$2.43
\$6.38 (Sep 2026)	\$0.98	\$2.74	\$2.59	\$2.50
\$7.00	\$1.08	\$2.85	\$2.70	\$2.61

Practice numbers: \$1,300 fixed a week, \$0.86 other variable costs a mile, 6.5 mpg, 9% fees. Print this table for your carrier and update it when costs change.

Per loaded mile vs per total mile

Load boards and rate tools show rates per **loaded** (trip) mile. Your break-even is per **total** mile. Convert before you compare:

Rate per total mile = all-in rate ÷ (deadhead + loaded miles)

Load 1: \$2,050 ÷ 760 loaded = \$2.70 “on the board”.
With 40 deadhead: \$2,050 ÷ 800 = **\$2.56**. That’s 3¢ under break-even.

The other direction: what should I ask for?

Ask = target × total miles. Load 1: \$2.70 × 800 = **\$2,160**. Walk-away (break-even): \$2.59 × 800 = **\$2,072**.



The \$2.40 story

Mike set \$2.40 in spring, when diesel was about \$5.00. Break-even then was **\$2.36**, so \$2.40 left 4¢. Diesel went to \$6.38 and break-even became \$2.59. Nobody updated the minimum, and Truck 12 started losing about 19¢ on every “minimum” mile.



Tell the carrier early

“Mike, with diesel at \$6.38 our break-even on Truck 12 is about \$2.59 a mile. Our \$2.40 minimum now loses money. Can we set a new minimum around \$2.70?”

Module 6 board, re-checked with today’s numbers

Load	All-in	Total mi	Per total mi	vs \$2.59 / \$2.70	Ask at \$2.70	Now
1 · Chicago → Charlotte	\$2,050	800	\$2.56	3¢ under break-even	\$2,160	COUNTER
7 · Kankakee → Jacksonville	\$2,700	1,085	\$2.49	10¢ under	\$2,930	COUNTER weak reload
2 · Bolingbrook → Atlanta	\$1,600	727	\$2.20	39¢ under	\$1,963	COUNTER check reload
8 · Gary → Atlanta (Thu)	\$1,950	740	\$2.64	above break-even	\$1,998	Backup: costs a day
3 · Joliet → Miami	\$3,200	1,370	\$2.34	25¢ under	\$3,699	PASS weak market

Load #12354 from your OMNITECH manual: Lebanon, OH → Owensboro, KY. Real document; numbers added for this course.

- 1 MILES 240: loaded miles only. Add your deadhead.
- 2 Flat rate \$700.00 and Total \$700.00: all-in, no separate fuel line. This is the number on your invoice.
- 3 \$150 deduction if the driver refuses or disables tracking
- 4 \$150 deduction if the POD is not sent within 72 hours

\$700 can become \$400. Two missed rules (tracking, POD) cost \$300, which is 43% of the pay. Read the notes before you book, and brief the driver (Module 9).

Money checklist for any rate con

1. All-in total, or linehaul + fuel?
2. Miles shown: loaded only?
3. Accessorials listed (detention, lumpers, TONU)?
4. Deductions and fines in the notes?
5. Payment terms: days, quick-pay fee, factor's address?



Factoring: commonly about 1.5–3% (recourse) or 2.5–5% (non-recourse) per invoice, plus possible transfer and same-day fees (FreightWaves).

Dispatch: often 5–10% of gross; 6–7% is common for vans (industry guides). Truck 12 uses 3% and 6%.

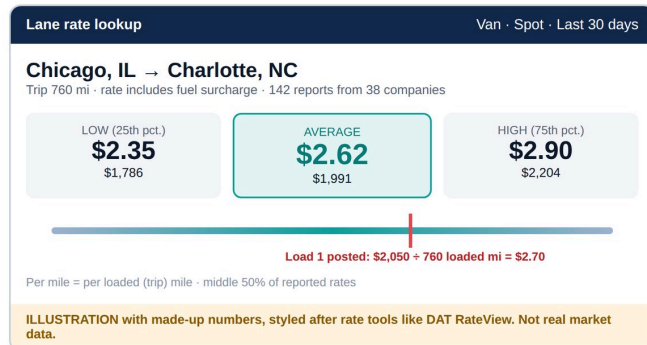
Follow Load 1's \$2,050 to the carrier's pocket

Step	Amount	Note
Rate confirmation, all-in	\$2,050.00	Chicago → Charlotte, 760 loaded + 40 deadhead
- Factoring 3%	- \$61.50	Paid within a day or two instead of waiting for the broker
- Dispatch 6%	- \$123.00	Your employer's fee (or your pay)
= Carrier keeps	\$1,865.50	91% of the rate
- Cost of 800 miles at \$2.36	- \$1,888.00	Fuel, driver, repairs + share of weekly fixed costs
= Result	- \$22.50	A small loss: why you counter toward \$2,160

What is the lane really paying?

Market rate tools: what the lane is really paying

Compare the posted rate with recent paid rates on the same lane, same equipment, before you call.



What real RateView shows (DAT help centre): rates from actual reported transactions (at least 7 records from 3 companies), the middle 50% range, spot (broker → carrier) for 7–90 days or a year, contract rates, and a 13-month history.

Fuel: single-lane lookups include fuel surcharge. DAT says to apply the **total** to your own miles, not their per-mile figure.

Use it three ways: Is the posted rate low, fair or high? Where should my counter-offer be? Is my break-even realistic on this lane?

How to read a lane range

- **Below the low end:** the posted rate is weak. Counter with confidence.
- **Around the average:** fair. Negotiate a little, decide on the whole trip.
- **Above the high end:** strong, or something is unusual (urgent load, hard pickup, fraud: check Module 8).

Rate tools usually need a paid subscription. Many carriers give their dispatcher access with the board seat.



Spot vs contract

Spot = one load, priced today, broker to carrier. **Contract** = rates agreed for months between shipper and carrier. You mostly work spot. DAT reports both, and they can move in different directions.



Don't quote from memory

Rates change every week. Never tell a carrier "Chicago to Charlotte pays \$2.60" from last month. Look it up again.

No rate tool? Four other ways to judge a rate

Method	How	Watch out
Compare similar posts	Search the same lane and day; note posted rates on 3–5 comparable loads	Many loads post without a rate
Your own history	Your load economics cards for that lane over the last weeks	Old weeks may not match today
Board market data	Many boards show lane averages or rate insights beside the load	Check the date range and whether fuel is included
Ask the broker	"What's your budget on this one?", then counter from your numbers (Module 8)	Their first number is rarely their best

Hot markets, cold markets

Supply and demand on the board

DAT Market Conditions and load-to-truck ratios show where trucks are scarce. Check them live; they change daily.



National snapshot (for context only)

DAT, August 2026	Spot linehaul	Fuel surcharge	≈ Spot all-in	Contract linehaul
Van	\$2.19	\$0.70	\$2.89	\$2.41
Reefer	\$2.61	\$0.77	\$3.38	\$2.65
Flatbed	\$2.70	\$0.84	\$3.54	—

Per loaded mile, national averages. "All-in" = linehaul + surcharge, added for this course. DAT called August's 20¢ van drop the steepest August pullback on record.



What this means for Truck 12

A national van average of about \$2.89 per loaded mile is only about **\$2.51 per total mile** once you add 15% deadhead. That's below Truck 12's \$2.59 break-even. In a soft market with expensive diesel, lane choice and negotiation decide whether the week makes money.



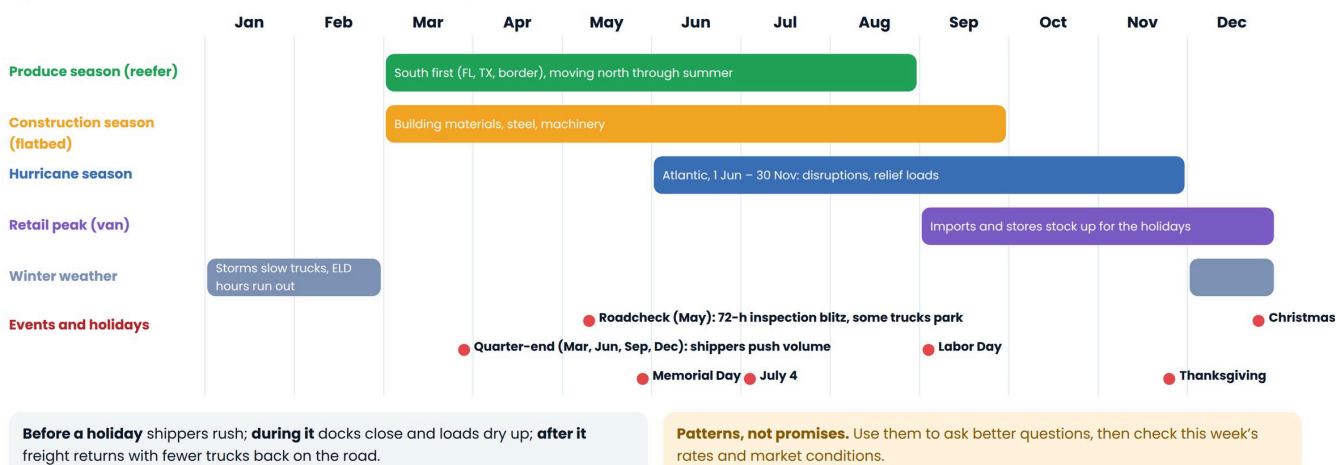
Averages hide lanes

Some lanes pay far above the national average and some far below. Use the national number to understand the market's mood. Use lane tools for real decisions.

The freight year

The freight year: seasonal patterns

Typical U.S. patterns that move demand and rates. Timing and strength change every year.



Questions seasons help you ask

- Is it produce season in the destination? (Reefers get scarce; van rates can rise too.)
- Is a holiday coming? (Plan pickups before docks close; expect a slow day after.)
- Is it quarter-end? (Shippers push volume in the last days.)
- Is weather coming? (Hurricane or winter storm: add time and ask about delays.)



Roadcheck 2026

CVSA's International Roadcheck ran **12–14 May 2026**: 72 hours of North American roadside inspections. Some drivers take those days off, so capacity can tighten briefly.



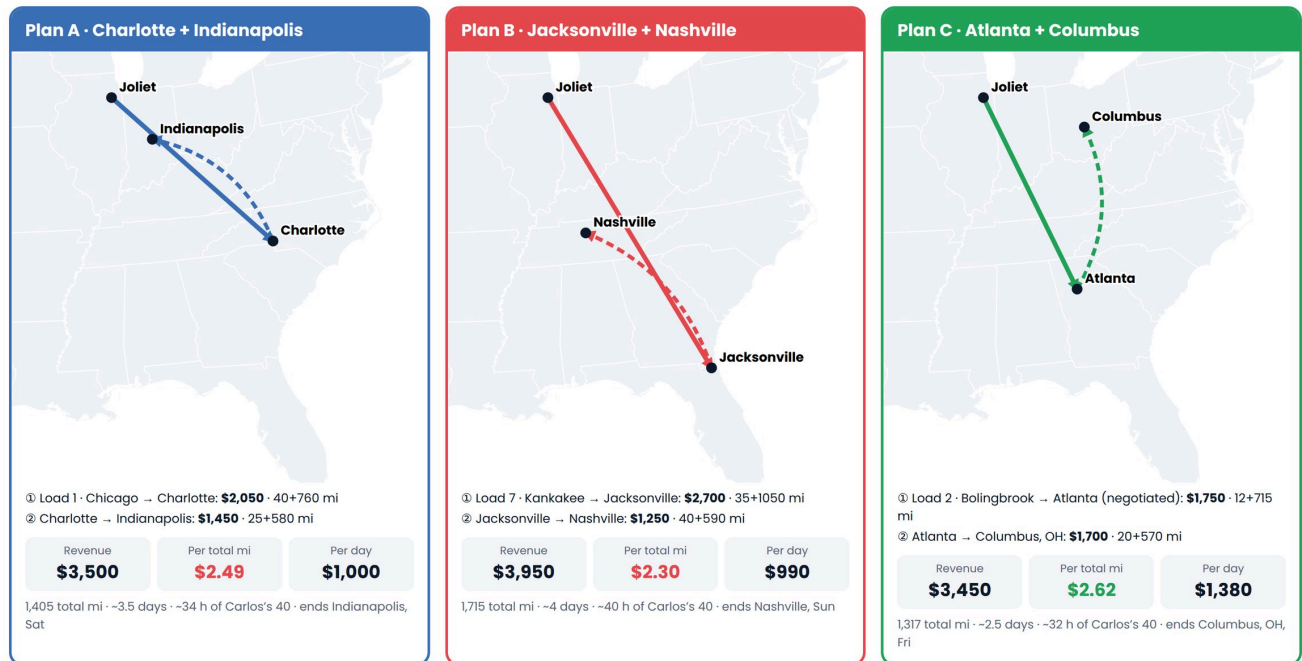
Weekly rhythm too

Mondays often bring fresh freight; Friday afternoons bring loads for Monday; weekends are thin in many markets. Watch your own board to learn your lanes.

Plan the week, not the load

Three ways to spend Truck 12's week

Practice lane plan from Joliet, Wed 7:30 AM. Break-even \$2.59 per total mile. Reloads and rates are practice values; maps drawn for this course.



All three plans start with Truck 12 empty in Joliet on Wednesday morning. Each has a first load from the Module 6 board and a practice reload. **Before you read the next page**, decide which plan you would pitch to Mike, and why.

Plan	Wednesday	Thursday	Friday	Weekend
A	Load 1, drive toward Charlotte	Deliver ~noon ET; wait	Reload Charlotte → Indianapolis	Deliver Sat
B	Load 7, drive south	Drive	Deliver Jacksonville; reload	Deliver Nashville Sat/Sun
C	Load 2, drive to Atlanta	Deliver AM; reload Atlanta PM	Deliver Columbus ~noon	Free for a short load or restart



A reload you found is not booked

Plan C depends on the Atlanta load still being there on Thursday. Ask the broker if you can book it now, or keep two backups. A plan is only as good as its weakest leg.



What the numbers don't show

Broker credit and days to pay (Module 6), appointment risk, weather, the driver's home time, and whether the carrier wants to be in that area. Ask Mike before you decide.

Per mile, per day, per week

Rate per mile vs rate per day

Same three plans, three different winners, depending on what you measure.

First load, per total mile

Plan A	\$2.56
Plan B	\$2.49
Plan C	\$2.41

B looks best here...

Whole trip, per total mile

Plan A	\$2.49
Plan B	\$2.30
Plan C	\$2.62

Break-even \$2.59: only C clears it

Revenue per day

Plan A	\$1,000
Plan B	\$990
Plan C	\$1,380

Truck 12 needs ≈ \$1,300 per working day

The best load is not always the best week. Plan B has the richest first load and the most revenue, but it ends deep in Florida-to-Tennessee freight, uses all of Carlos's hours and earns the least per day.

Plan C starts with the weakest load (\$2.41) but Atlanta's reload pays well, the truck finishes Friday near the Midwest, and Carlos keeps 8 hours for one more short load.

Rate per day

Revenue per day = trip revenue ÷ days the truck is committed, from dispatch until it is empty again and ready.

Truck 12 needs about **\$6,500 a week** (2,500 mi × \$2.59) to break even, which is about **\$1,300 per working day** over five days.

Plan	Margin vs break-even	Hours used
A	-\$144	~34 of 40
B	-\$498	~40 of 40
C	+\$35 and 8 h left for a short load	~32 of 40

Margin = revenue - (total miles × \$2.59).



Correction: "Always take the highest rate per mile"

Short loads with high per-mile rates can pay less per day. Long loads into weak markets can look rich and still lose the week. Measure **per total mile**, **per day** and **where the truck ends up**.



Hours are money too

Plan B uses all 40 of Carlos's remaining hours (Module 4). Plan C leaves room for one more short load before a 34-hour restart.

From a number to a recommendation

From a number to a decision

Five checks, about a minute per load, before you recommend it to the carrier.



The load economics card

A one-minute worksheet. Print it or copy it into your notes app.

Load economics card • Load ID _____ • Broker _____		your numbers
Posted / offered rate (all-in)		\$
Loaded miles		mi
Deadhead to pickup		mi
Total miles		mi
Rate per total mile		\$ / mi
Carrier break-even • target		\$ • \$
Market range for lane (low-avg-high)		\$ - \$ - \$
Days used (dispatch → empty again)		days
Revenue per day		\$ / day
Next-leg market: strong / OK / weak		
Hours check (Module 4)		OK / tight / no
Decision: BOOK • COUNTER • PASS		

Fill one card per serious load. It takes a minute and turns a gut feeling into numbers the carrier can check.

Worked: Load 1
 $\$2,050 \cdot 760 + 40 = 800 \text{ mi} \cdot \$2.56 \cdot \text{BE } \$2.59 / \text{target}$
 $\$2.70 \cdot \sim 1.5 \text{ days} \approx \$1,370/\text{day} \cdot \text{next leg OK} \cdot \text{hours OK}$
 → **COUNTER** (ask $\approx \$2,160 = \2.70)

Keep the cards: after a month they show the carrier's real lanes, rates and which brokers pay.

Pitch the plan to the carrier

What a good pitch contains

1. **The load:** lane, miles, pickup and delivery.
2. **The money:** offered rate, your counter, rate per total mile.
3. **The comparison:** against break-even and the market.
4. **The reason:** the next leg, days, hours, where the truck ends.
5. **The question:** "Do you want me to book it?"



The carrier decides

Never book below the carrier's minimum without asking. Some carriers let you book above target on your own: get that permission in writing (Module 1 dispatch agreement).



Say it out loud • drill 9: pitching a plan

"Mike, I have Bolingbrook to Atlanta, 715 miles, 12 empty, picks up Wednesday at 9, delivers Thursday at 1. The broker offered \$1,600. I got them to \$1,750, which is \$2.41 a mile, under our \$2.59 break-even by itself. But Atlanta is strong: I have a Thursday reload to Columbus for \$1,700, so the two loads average \$2.62, and Carlos is back near home Friday with hours left. Do you want me to book both?"



Your turn

Write and say the pitch for **Plan A**. Be honest that the trip averages \$2.49, 10¢ under break-even, and suggest what you would try to improve it.

Words that make money talk clear

Say	Instead of
"\$2.56 per total mile"	"good money" / "decent rate"
"3 cents under break-even"	"a bit low"
"\$1,380 a day for 2.5 days"	"quick load"
"Atlanta shows many outbound loads today"	"Atlanta is always hot"

Money drills

Use a calculator, write your working, then check the answer key. Round money to the cent, rates to the cent per mile.

1 • Build a cost per mile (reefer)

A 53' reefer: fixed costs **\$1,500 a week**, runs **2,800 total miles** a week. Diesel **\$6.38, 6.0 mpg**. Driver **\$0.62**, repairs **\$0.20**, tyres **\$0.05**, tolls **\$0.03**, reefer unit fuel and service **\$0.10** a mile. Dispatch 5%, factoring 3%.

Find	Answer
Fixed per mile	
Variable per mile	
Operating cost per mile	
Break-even per total mile	

2 • What if...? (Truck 12)

- a) Diesel falls to **\$5.50**. New break-even?
 b) Diesel stays \$6.38 but the truck runs only **2,000 miles** a week. New break-even?

3 • Fuel surcharge

A broker offers **\$1,400 linehaul + FSC** for **700 loaded miles**, DAT method (base \$1.25, 6.0 mpg), diesel **\$6.38**. Deadhead 50 miles.

Find	Answer
FSC per mile	
FSC total	
All-in total	
Per total mile vs Truck 12's \$2.59	

4 • Per mile or per day?

Load X: \$1,100, 300 loaded + 20 deadhead, takes 1 day.

Load Y: \$2,400, 950 loaded + 50 deadhead, takes 2 days.

Work out per total mile and per day for each.

Which would you pick, and what would change your mind?

5 • Rate con deductions

On load #12354 (\$700), the driver turns off tracking and the POD is sent on day 4. What is the carrier paid? What percentage was lost?

Speed round: rate per total mile in your head (target: 10 seconds each)

a	b	c	d	e	f
\$1,800 · 750 mi	\$2,500 · 1,000 mi	\$1,350 · 540 mi	\$3,100 · 1,150 mi	\$950 · 410 mi	\$2,200 · 880 mi

Answers: a \$2.40 · b \$2.50 · c \$2.50 · d \$2.70 · e \$2.32 · f \$2.50. Tip: round the miles, divide, then adjust.

Plan Truck 12's week

Use the three plans on page 14. Fill the table without looking at page 15, then rank the plans and write your pitch.

Measure	Plan A · Charlotte + Indianapolis	Plan B · Jacksonville + Nashville	Plan C · Atlanta + Columbus
Revenue			
Total miles			
Per total mile			
vs break-even \$2.59			
Days · per day			
Hours used of 40			
Where the truck ends			
Rank (1–3)			

6 · Improve Plan A

Plan A averages \$2.49. Give **two** ways a dispatcher could lift it above break-even. (Hint: the first load's price, the second load's price, or the idle time on Thursday.)

7 · Explain it to Mike

Mike asks: “**Why can’t we just keep \$2.40 as our minimum?**” Write a three-sentence answer with numbers.



Speaking practice

Record yourself saying drill 9 for your top-ranked plan, in under 45 seconds. Listen back: are the numbers clear? Did you end with a question?

Module 7 check

1. What is cost per mile? Why do we divide by total miles, not loaded miles?
2. Name three fixed costs and three variable costs.
3. Why does fixed cost per mile fall when the truck runs more miles?
4. Diesel rises from \$5.00 to \$6.38. How much does Truck 12's fuel cost per mile change (6.5 mpg)?
5. Write the break-even formula. Why divide by (1 - fee %) instead of adding the fees?
6. What are Truck 12's operating cost, break-even and target minimum?
7. Calculate DAT's fuel surcharge at \$6.00 diesel and 6.0 mpg.
8. What is the difference between an all-in rate and linehaul + FSC?
9. What does a rate tool's low-average-high range show? What minimum data does DAT RateView need to show a lane?
10. Define load-to-truck ratio. Why compare a market with its own history?
11. Give two reasons a load with a lower rate per mile can be the better choice.
12. What two deductions are on the Ship Express rate con, and how much money are they?

Skills self-check: tick when you can do it without notes

Skill	Can do	Need practice
Build a carrier's cost per mile from fixed and variable costs	☐	☐
Calculate break-even and a target minimum with percentage fees	☐	☐
Update the break-even when diesel changes	☐	☐
Convert linehaul + FSC to all-in, and loaded to total miles	☐	☐
Compare a posted rate with a lane range	☐	☐
Compare plans per total mile, per day and by end location	☐	☐
Pitch a plan to the carrier with numbers (drill 9)	☐	☐

Answers: practice

- 1** Fixed $\$1,500 \div 2,800 = \0.54 . Fuel $\$6.38 \div 6.0 = \1.06 ; variable = $1.06 + 0.62 + 0.20 + 0.05 + 0.03 + 0.10 = \2.06 . Operating = **\$2.60**. Break-even = $\$2.60 \div 0.92 = \2.83 .
- 2a** Fuel $\$5.50 \div 6.5 = \0.85 . Cost = $0.52 + 0.85 + 0.86 = \$2.23$. $\div 0.91 = \$2.45$.
- 2b** Fixed $\$1,300 \div 2,000 = \0.65 . Cost = $0.65 + 1.84 = \$2.49$. $\div 0.91 = \$2.74$.
- 3** FSC = $(6.38 - 1.25) \div 6.0 = \$0.855/\text{mi.} \times 700 = \599 . All-in = **\$1,999**. $\div 750$ total mi = **\$2.67**, 8¢ above break-even.
- 4** X: $\$1,100 \div 320 = \$3.44/\text{total mi, } \$1,100/\text{day}$. Y: $\$2,400 \div 1,000 = \$2.40/\text{total mi, } \$1,200/\text{day}$. Y earns more per day but is below break-even per mile. X is better only if a good reload is ready the next morning; ask about both destinations before choosing.
- 5** $\$700 - \$150 - \$150 = \400 . Lost $\$300 = 43\%$.

	A	B	C
Revenue	\$3,500	\$3,950	\$3,450
Total mi	1,405	1,715	1,317
Per total mi	\$2.49	\$2.30	\$2.62
vs \$2.59	-\$144	-\$498	+\$35
Days · per day	3.5 · \$1,000	4 · \$990	2.5 · \$1,380
Hours	~34	~40	~32
Ends	Indianapolis, Sat	Nashville, Sun	Columbus, Fri
Rank	2	3	1

- 6** Any two: counter Load 1 toward \$2,160 (+\$110 lifts the trip to \$2.57); negotiate the Indianapolis reload up by \$150 (with the Load 1 counter too: $\$3,760 = \2.68); find a Thursday reload instead of Friday to save a day; pick a reload that ends closer to Joliet to cut next week's deadhead.
- 7** Example: "At \$6.38 diesel Truck 12 costs about \$2.36 a mile, and after 9% fees we need \$2.59 just to break even. At \$2.40 we lose about 19¢ a mile, around \$475 on a 2,500-mile week. I suggest a \$2.70 minimum and we review it whenever diesel moves 25¢."

Answers: module check

- 1 All costs ÷ all miles for a period. Empty miles cost fuel, pay and wear too, so leaving them out makes the truck look cheaper than it is.
- 2 Fixed: truck payment, trailer payment, insurance, ELD/phone/board seat, permits. Variable: fuel, driver pay per mile, repairs, tyres, tolls.
- 3 The weekly fixed amount is shared across more miles: $\$1,300 \div 2,000 = \0.65 , $\div 3,000 = \$0.43$.
- 4 $\$5.00 \div 6.5 = \$0.77 \rightarrow \$6.38 \div 6.5 = \0.98 : **+21¢** a mile.
- 5 Break-even = cost per mile ÷ (1 - fee %). Fees are a percentage of the rate, and the rate is the unknown. Adding the fees to the cost undercounts them.
- 6 $\$2.36 \cdot \$2.59 \cdot \$2.70$ (practice, diesel \$6.38).
- 7 $(\$6.00 - \$1.25) \div 6.0 = \mathbf{\$0.79}$ a mile.
- 8 All-in = one total that already includes fuel. Linehaul + FSC = base pay plus a separate fuel amount; convert to all-in to compare.
- 9 The spread of recently reported paid rates (middle 50%) around the average. RateView needs at least 7 records from 3 different companies.
- 10 Load posts ÷ truck posts for one market and equipment type. Levels differ by market and equipment (flatbed runs much higher than van), so "high" only means high compared with that market's normal.
- 11 It leads to a strong reload or good end location; it uses fewer days (more per day); it fits the driver's hours; it avoids a weak market.
- 12 \$150 if tracking is refused or disabled; \$150 if POD is not sent within 72 hours. \$300 total.



Try-it answer (page 17)

Plan A pitch example: "Mike, Plan A is Load 1 to Charlotte for \$2,050, then Charlotte to Indianapolis Friday for \$1,450. Together that's \$3,500 on 1,405 miles, \$2.49 a mile, about 10¢ under break-even. I'd try to get Load 1 up to \$2,160 and look for a Thursday reload. Do you want me to try?"

Glossary

All-in rate — One total that includes fuel.

Break-even — Rate per total mile where revenue after fees equals cost.

Contract rate — Rate agreed for months between shipper and carrier.

Cost per mile — All costs ÷ all miles in a period.

Deduction — Money taken off the pay for a broken rule.

Dispatch fee — Dispatcher's pay, often a % of gross.

EIA — U.S. Energy Information Administration; publishes weekly diesel prices.

Factoring — Selling invoices for fast pay, for a fee (recourse / non-recourse).

Fixed cost — Paid every week regardless of miles.

Fuel surcharge (FSC) — Pay that follows the diesel price.

Headhaul / backhaul — Strong direction / weaker return direction.

Linehaul — Base rate without fuel.

Load-to-truck ratio — Load posts ÷ truck posts, per market and equipment.

Market Conditions — DAT's hot/cold supply-demand map.

Operating cost — Fixed + variable cost per mile.

Rate per day — Trip revenue ÷ days the truck is committed.

Rate per total mile — All-in rate ÷ (deadhead + loaded).

RateView — DAT's tool for recent paid lane rates.

Spot rate — One-off price for a single load.

Target minimum — Break-even plus the carrier's profit cushion.

Variable cost — Grows with each mile.

Walk-away rate — Lowest rate you'd accept: usually break-even.

Formula card

What	Formula	Truck 12
Fixed per mile	weekly fixed ÷ weekly total miles	\$1,300 ÷ 2,500 = \$0.52
Fuel per mile	diesel price ÷ mpg	\$6.38 ÷ 6.5 = \$0.98
Operating cost	fixed per mile + variable per mile	\$2.36
Break-even	operating cost ÷ (1 - fee %)	\$2.36 ÷ 0.91 = \$2.59
Target minimum	(cost + profit per mile) ÷ (1 - fee %)	\$2.46 ÷ 0.91 = \$2.70
Fuel surcharge (DAT method)	(EIA diesel - \$1.25) ÷ mpg	(\$6.38 - 1.25) ÷ 6 = \$0.86
Rate per total mile	all-in ÷ (deadhead + loaded)	\$2,050 ÷ 800 = \$2.56
Ask price	target × total miles	\$2.70 × 800 = \$2,160
Revenue per day	trip revenue ÷ days committed	\$3,450 ÷ 2.5 = \$1,380

Sources

Source	What it supports	Where
ATRI	An Analysis of the Operational Costs of Trucking, 2026 update (2025 data): \$2.336/mi, \$1.854 non-fuel	truckingresearch.org
HDT / Trucking Info	ATRI line items: driver wages 81.8¢, benefits 21¢, repairs 21.5¢, tyres 5¢; margins	truckinginfo.com
EIA	Gasoline and Diesel Fuel Update: \$6.382 (28 Sep 2026), \$3.754 a year earlier	eia.gov/petroleum/gasdiesel
DAT iQ	Fuel surcharge calculation in RateView: $(\text{EIA} - \$1.25) \div \text{MPG}$	iq.support.dat.com
DAT iQ	RateView FAQs: transaction data, 7 records / 3 companies, middle 50%, spot/contract periods, fuel inclusion	iq.support.dat.com
DAT	August 2026 national rates and surcharges (news release, 15 Sep 2026)	dat.com · globenewswire.com
DAT	What is the load-to-truck ratio? · Using Market Conditions (DAT iQ)	dat.com/blog · iq.support.dat.com
CVSA	International Roadcheck, 12–14 May 2026	cvsa.org
FreightWaves	Freight factoring rates: recourse 1.5–3%, non-recourse 2.5–5%, hidden fees	freightwaves.com
O Trucking	Dispatcher fees: 5–10%, 6–7% common for van/reefer (industry guide)	otrucking.com
User-supplied	OMNITECH manual: Ship Express rate confirmation #12354; route-map screenshot	your files



Practice values

Truck 12's costs, the \$2.70 target, the rate-tool card and all reloads in Plans A–C are practice values made for this course. Seasonal patterns are general industry knowledge, not forecasts.



Recheck before real use

Diesel prices, DAT averages, factoring and dispatch fee ranges change. Check the current source before quoting a number to a carrier.